ECON2050 Tutorial 3

Basic Real Analysis III

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Today's Plan

- Assignment help
- Tutorial questions: Q1–Q4
- QA

Assignment Q1

Question 1

Consider the following sequences. For each of the sequences, determine whether it converges or diverges. If the sequence converges, prove which number or vector is the limit of the sequence. If the sequence diverges, carefully explain and justify why.

(a) $\left(10 + \frac{1}{n}\right)_{n \in \mathbb{N}}$ (5 marks)

(b) $\left(\left(5 - \frac{1}{n^4}\right)\left(-\frac{1}{3}\right)^n\right)_{n \in \mathbb{N}}$ (10 marks)

(c) $\left(\frac{n^2+5n+6}{n^2}, \frac{1}{n}\cos(n)\right)_{n \in \mathbb{N}}$ (15 marks)

Assignment Q1

Sequences in $\mathbb{R}$

A sequence (x_n) with codomain $\mathbb{R}$ converges to $x^* \in \mathbb{R}$ if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N : |x_n - x^*| < \varepsilon.$$

Sequences in $\mathbb{R}^m$

A sequence $(\boldsymbol{x}_n)$ with codomain $\mathbb{R}^m$ converges to $\boldsymbol{x}^* \in \mathbb{R}^m$ if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N : \|\boldsymbol{x}_n - \boldsymbol{x}^*\| < \varepsilon.$$

Componentwise convergence

For $\boldsymbol{x}_n = (x_{1,n}, \ldots, x_{m,n}) \in \mathbb{R}^m$,

$$\boldsymbol{x}_n \to \boldsymbol{x}^* \quad \Longleftrightarrow \quad x_{i,n} \to x_i^* \text{ for every } i = 1, \ldots, m.$$

Assignment Q2

Question 2

Draw the following sets. Explain whether they are open, closed, both or neither. Define the set of interior points and boundary points. Carefully justify whether the set is or is not compact.

(a) $S_1 = \{(x, y) \in \mathbb{R}^2 : (x+2)^2 + (y+2)^2 \leq 4,\ y \leq x\}$ (5 marks)

(b) $S_2 = \{(x, y) \in \mathbb{R}^2 : y \leq 2\sin(3x),\ y \geq 1\}$ (5 marks)

(c) $S_3 = \{(x, y) \in \mathbb{R}^2 : y \leq -|x| + 3,\ y \geq \cos(x)\}$ (5 marks)

Assignment Q2

Open and closed

A point $\boldsymbol{x} \in \mathbb{R}^m$ is

- an **interior point** of S if some open ball centered at $\boldsymbol{x}$ is contained entirely in S;
- an **exterior point** of S if some open ball centered at $\boldsymbol{x}$ is contained entirely outside S;
- a **boundary point** of S if every open ball centered at $\boldsymbol{x}$ contains both a point in S and a point outside S.

A set $S \subseteq \mathbb{R}^m$ is

- **open** if all of its points are interior points;
- **closed** if it contains all of its boundary points.

Bounded and compact

A set $S \subseteq \mathbb{R}^m$ is bounded if

$$\exists K > 0 \text{ s.t. } \forall \boldsymbol{x} \in S : \|\boldsymbol{x}\| \leq K.$$

A set is **compact** if it is closed and bounded.

Assignment Q3

Question 3

(a) Prove that

$$f : \mathbb{R}^2 \to \mathbb{R}, \ (x_1, x_2) \mapsto \sin\left(\sqrt{x_1^2(\cos(x_2))^4}\right)$$

is continuous. For this question, you may refer to theorems and results from the lecture slides without proof. (10 marks)

(b) Consider:

$$f : \mathbb{R}^2 \setminus \{(x_1, x_2) \in \mathbb{R}^2 : x_1 + x_2^2 = 0\} \to \mathbb{R}, \quad (x_1, x_2) \mapsto \frac{x_1 x_2^4}{x_1^2 + x_2^6}$$

Prove that every function $g : \mathbb{R}^2 \to \mathbb{R}$ such that $f(x_1, x_2) = g(x_1, x_2)$ for all

$$(x_1, x_2) \in \mathbb{R}^2 \setminus \{(x_1, x_2) \in \mathbb{R}^2 : x_1 + x_2^2 = 0\}$$

is discontinuous at $(0, 0)$. (10 marks)

Assignment Q3

Tools for Proving Continuity

- Algebra of continuous functions
- Composition function
- Componentwise continuity
- Projection functions

Sequential definition

Let $D \subseteq \mathbb{R}^k$. A function $f : D \to \mathbb{R}^m$ is **continuous at** $\boldsymbol{x} \in D$ if for every sequence $(\boldsymbol{x}_n)$ in D such that $\boldsymbol{x}_n \to \boldsymbol{x}$, we have $f(\boldsymbol{x}_n) \to f(\boldsymbol{x})$.

An example: page 14, lecture 2.

Assignment Q4

Question 4

Prove the triangle inequality. That is,

$$\text{Let } x, y \in \mathbb{R}, \text{ then } |x + y| \leq |x| + |y|$$

Note: Other properties of the absolute value function can be used, so long as it is communicated clearly what they are, why and how they are used here.

Hint

Try to show $-|t| \leq t \leq |t|$.

Tutorial Q1

Question 1

(a) Let

$$f\colon \mathbb{R}^3_{++} \to \mathbb{R}, \qquad (x_1, x_2, x_3) \mapsto \frac{3x_1^2 + e^{x_2}}{2x_3}.$$

Compute all partial derivatives.

(b) Evaluate the partial derivative of f with respect to x_3 at the point $(1, 2, 3)$.

Recap: Partial Derivatives

Definition

Let $f\colon D \to \mathbb{R}$ with $D \subseteq \mathbb{R}^k$ open. The partial derivative with respect to x_i is

$$\frac{\partial f(\boldsymbol{x})}{\partial x_i}.$$

How to calculate

Differentiate with respect to the variable of interest while holding all other variables constant.

Tutorial Q1: Partial Derivatives

Knowledge used

- Use the lecture definition of partial derivatives.
- Hold all other variables constant and apply one-variable differentiation rules.
- Use the chain rule for e^{x_2} and the power rule for x_3^{-1}.

Solution

$$f(x_1, x_2, x_3) = \frac{3x_1^2 + e^{x_2}}{2x_3}.$$

$$\frac{\partial f(x_1, x_2, x_3)}{\partial x_1} = \frac{3x_1}{x_3}, \qquad \frac{\partial f(x_1, x_2, x_3)}{\partial x_2} = \frac{e^{x_2}}{2x_3},$$

$$\frac{\partial f(x_1, x_2, x_3)}{\partial x_3} = -\frac{3x_1^2 + e^{x_2}}{2x_3^2}.$$

Tutorial Q1(b): Evaluation

Knowledge used

- First compute the partial derivative function.
- Then substitute the point $(1, 2, 3)$ into that partial derivative.

Solution

From part (a),

$$\frac{\partial f(x_1, x_2, x_3)}{\partial x_3} = -\frac{3x_1^2 + e^{x_2}}{2x_3^2}.$$

Therefore,

$$\frac{\partial f(1, 2, 3)}{\partial x_3} = -\frac{3 \cdot 1^2 + e^2}{2 \cdot 3^2} = -\frac{3 + e^2}{18}.$$

Tutorial Q2

Question 2

You are given

$$f\colon D \to \mathbb{R}, \qquad (x, y) \mapsto e^{x/y} + \ln(x^2 + 1),$$

where D is the largest subset of $\mathbb{R}^2$ on which f is well-defined.

(a) Determine D.

(b) Calculate the gradient of f at every point (x, y).

(c) Show that all points of the form $(0, y)$, $y \neq 0$, are located on the same level curve.

(d) Determine, without checking the definition directly, whether $f(x, y)$ is continuously differentiable at $(1, 1)$.

Recap: Gradient and Continuously Differentiability

Gradient

The gradient of f at $\boldsymbol{x}_0$ is the vector of partial derivatives:

$$\nabla f(\boldsymbol{x}_0) = \left(\frac{\partial f(\boldsymbol{x}_0)}{\partial x_1}, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_2}, \ldots, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_k} \right).$$

Level curve

The level curve of level c is $\{\boldsymbol{x} \in D : f(\boldsymbol{x}) = c\}$.

Continuously differentiability

If a funtion is differentiable, the derivative is $Df(\boldsymbol{x}) \equiv \boldsymbol{a}$. If $\boldsymbol{x} \mapsto Df(\boldsymbol{x})$ is continuous, then f is **continuously differentiable**.

Useful theorems

If f is continuously partially differentiable, then f is continuously differentiable.

Tutorial Q2(a): Domain

Knowledge used

- The domain is the largest set where the formula is well-defined.
- Check every operation in the formula: division, exponentials and logarithms.

Solution

The function is

$$f(x, y) = \mathrm{e}^{x/y} + \ln(x^2 + 1).$$

The term $\mathrm{e}^{x/y}$ requires $y \neq 0$.
The term $\ln(x^2 + 1)$ is always well-defined because $x^2 + 1 > 0$ for all $x \in \mathbb{R}$.
Therefore,

$$D = \{(x, y) \in \mathbb{R}^2 : y \neq 0\}.$$

Tutorial Q2(b): Gradient

Knowledge used

- The gradient is the vector of partial derivatives.
- Use the chain rule and hold the other variable fixed.

Solution

The partial derivatives are

$$\frac{\partial f(x,y)}{\partial x} = \frac{1}{y}\mathrm{e}^{x/y} + \frac{2x}{x^2+1}, \qquad \frac{\partial f(x,y)}{\partial y} = -\frac{x}{y^2}\mathrm{e}^{x/y}.$$

Therefore,

$$\nabla f(x,y) = \left(\frac{1}{y}\mathrm{e}^{x/y} + \frac{2x}{x^2+1}, -\frac{x}{y^2}\mathrm{e}^{x/y}\right).$$

Tutorial Q2(c): Level Curve

Knowledge used

- A level curve of level c is the set of points in the domain satisfying $f(x, y) = c$.
- To show points are on the same level curve, show that f takes the same value at those points.

Solution

For every point $(0, y)$ with $y \neq 0$,

$$f(0, y) = \mathrm{e}^{0/y} + \ln(0^2 + 1) = \mathrm{e}^0 + \ln(1) = 1 + 0 = 1.$$

Thus $f(0, y)$ does not depend on y.
Therefore, all points of the form $(0, y)$ with $y \neq 0$ are on the same level curve:

$$f(x, y) = 1.$$

Tutorial Q2(d): Continuous Differentiability

Knowledge used

- Continuous partial differentiability implies continuous differentiability.
- Use continuity tools: sums, products, quotients and compositions of continuous functions are continuous where defined.

Solution

Around $(1,1)$, $y \neq 0$, so both partial derivative functions are well-defined:

$$\frac{\partial f(x,y)}{\partial x} = \frac{1}{y}e^{x/y} + \frac{2x}{x^2+1}, \qquad \frac{\partial f(x,y)}{\partial y} = -\frac{x}{y^2}e^{x/y}.$$

Both expressions are built from continuous functions using sums, products, quotients and compositions, and the denominators are nonzero near $(1,1)$.
Hence both partial derivatives are continuous at $(1,1)$. Therefore,

f is continuously differentiable at $(1,1)$.

Tutorial Q3

Question 3

Consider the function

$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad (x, y) \mapsto \frac{xy}{x^2 + y^2 + 1}.$$

Determine

$$\frac{\partial f(1,1)}{\partial x}$$

directly from the definition of the partial derivative, that is, by calculating the appropriate limit.

Tutorial Q3

Question 3

Consider the function

$$f\colon \mathbb{R}^2 \to \mathbb{R}, \qquad (x, y) \mapsto \frac{xy}{x^2 + y^2 + 1}.$$

Determine

$$\frac{\partial f(1,1)}{\partial x}$$

directly from the definition of the partial derivative, that is, by calculating the appropriate limit.

Definition

Let $f\colon D \to \mathbb{R}$ with $D \subseteq \mathbb{R}^k$ open. The partial derivative with respect to x_i is

$$\frac{\partial f(\boldsymbol{x})}{\partial x_i} = \lim_{n\to\infty} \frac{f(x_1, \ldots, x_i + h_n, \ldots, x_k) - f(x_1, \ldots, x_i, \ldots, x_k)}{h_n},$$

provided this limit exists and is the same finite real number for all sequences $h_n \to 0$ with $h_n \neq 0$.

Tutorial Q3: Directly from the Definition

Knowledge used

- Use the definition of the partial derivative with respect to x.
- Hold $y = 1$ fixed and let the x-coordinate move from 1 to $1 + h_n$.

Solution

$$\frac{\partial f(1,1)}{\partial x} = \lim_{n\to\infty} \frac{f(1+h_n,1) - f(1,1)}{h_n}.$$

Since $f(1,1) = 1/3$,

$$\begin{aligned}\frac{\partial f(1,1)}{\partial x} &= \lim_{n\to\infty} \frac{\frac{1+h_n}{2+(1+h_n)^2} - \frac{1}{3}}{h_n} \\ &= \lim_{n\to\infty} \frac{3+3h_n-2-(1+h_n)^2}{3h_n(2+(1+h_n)^2)} \\ &= \lim_{n\to\infty} \frac{h_n(1-h_n)}{3h_n(2+(1+h_n)^2)} = \lim_{n\to\infty} \frac{1-h_n}{3(2+(1+h_n)^2)} = \frac{1}{9}.\end{aligned}$$

Tutorial Q4

Question 4

The production function of a firm is

$$f : \mathbb{R}^2_+ \to \mathbb{R}, \qquad (K, L) \mapsto 4K^{3/4}L^{1/4}.$$

Currently, the firm is using $10,000$ units of capital and employing 625 workers. The CEO would like to know how production changes when capital is increased by one unit and one more worker is employed. Can you help her?

Recap: Derivative Approximation

Gradient

The gradient of f at $\boldsymbol{x}_0$ is the vector of partial derivatives:

$$\boldsymbol{\nabla} f(\boldsymbol{x}_0) = \left(\frac{\partial f(\boldsymbol{x}_0)}{\partial x_1}, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_2}, \ldots, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_k} \right).$$

First-order approximation

For $\Delta \boldsymbol{x}$ sufficiently close to $\mathbf{0}$,

$$\Delta y = f(\boldsymbol{x}_0 + \Delta \boldsymbol{x}) - f(\boldsymbol{x}_0) \approx \mathrm{d}y = \boldsymbol{\nabla} f(\boldsymbol{x}_0) \cdot \Delta \boldsymbol{x}.$$

Equivalently,

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}_0) + \boldsymbol{\nabla} f(\boldsymbol{x}_0) \cdot (\boldsymbol{x} - \boldsymbol{x}_0).$$

Tutorial Q4: Derivative Approximation

Knowledge used

- Use partial derivatives as marginal products.
- Use the first-order approximation

$$\Delta f \approx \mathrm{d}f = \frac{\partial f}{\partial K}\mathrm{d}K + \frac{\partial f}{\partial L}\mathrm{d}L.$$

Solution

First compute

$$f(10000, 625) = 4(10000)^{3/4}(625)^{1/4} = 20000.$$

The partial derivatives are

$$\frac{\partial f(K,L)}{\partial K} = 4 \cdot \frac{3}{4}K^{-1/4}L^{1/4} = \frac{3}{4}\frac{f(K,L)}{K}, \qquad \frac{\partial f(K,L)}{\partial L} = 4 \cdot \frac{1}{4}K^{3/4}L^{-3/4} = \frac{1}{4}\frac{f(K,L)}{L}.$$

Tutorial Q4: Change in Output

Knowledge used

- The tangent hyperplane gives a local linear approximation to the production function.
- The approximation is useful when the change in inputs is small.

Solution

At $(K, L) = (10000, 625)$,

$$\frac{\partial f}{\partial K} = \frac{3}{4}\frac{20000}{10000} = \frac{3}{2}, \qquad \frac{\partial f}{\partial L} = \frac{1}{4}\frac{20000}{625} = 8.$$

Using $\mathrm{d}K = 1$ and $\mathrm{d}L = 1$,

$$\Delta f \approx \mathrm{d}f = \frac{3}{2} \cdot 1 + 8 \cdot 1 = 9.5.$$

Thus output increases by approximately 9.5 units.
Exact check:

$$f(10001, 626) = 4(10001)^{3/4}(626)^{1/4} \approx 20009.4958,$$

so the exact increase is approximately 9.4958 units.

Key Takeaways

- Partial derivatives measure the effect of changing one variable while holding others fixed.
- The gradient collects all partial derivatives into a vector.
- If a function is continuously partially differentiable, we can use the gradient for first-order approximations.